EXPERIMENT3

Aim : To Perform various GIT operations on local and Remote repositories using GIT Cheat-Sheet

Pull and Push Processes

The pull command used to fetch the repository from github to local while push is used to commit files from local repository to Github.

Push□ Push changes to Web repository Pull □

Pull changes to Local repository

The following commands are used for pull and push repositories

A) Push command

```
PS C:\Users\HP 15L\Desktop\T21-79> git init
Initialized empty Git repository in C:/Users/HP 15L/Desktop/T21-79/.git/
PS C:\Users\HP 15L\Desktop\T21-79>
```

```
$ git remote add origin https://github.com/bhushanjadhav1/siesworkshop.git
```

```
Initialized empty Git repository in C:/Users/HP 15L/Desktop/T21-79/.git/
PS C:\Users\HP 15L\Desktop\T21-79> git remote add origin https://github.com/tsecprac
tical-star/SEPM_PRAC.git
PS C:\Users\HP 15L\Desktop\T21-79>
```

```
$ git remote show origin
```

```
PS C:\Users\HP 15L\Desktop\T21-79> git remote show origin
* remote origin
  Fetch URL: https://github.com/tsecpractical-star/SEPM_PRAC.git
  Push  URL: https://github.com/tsecpractical-star/SEPM_PRAC.git
  HEAD branch: main
  Remote branch:
    main new (next fetch will store in remotes/origin)
PS C:\Users\HP 15L\Desktop\T21-79>
```

```
PS C:\Users\HP 15L\Desktop\T21-79> git commit -m "initial commit"
[master (root-commit) 0b77317] initial commit
 1 file changed, 10 insertions(+)
 create mode 100644 exp5.py
PS C:\Users\HP 15L\Desktop\T21-79> git push -u origin main
error: src refspec main does not match any
```

If you add remote again then will show you fatal error.

$ git remote add origin

https://github.com/bhushanjadhav1/Myrepository.git fatal: remote origin

already exists. So, to delete origin rm origin command is used

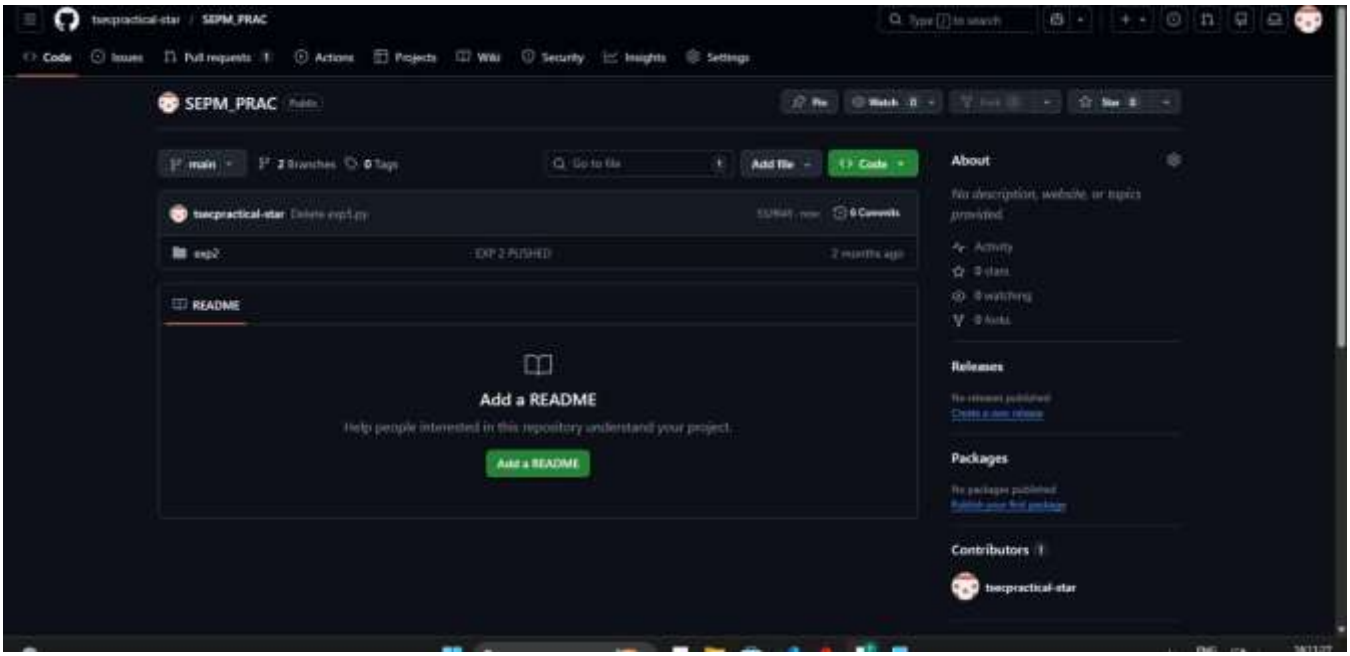
$ git remote rm origin

Now, to push the local repository to remote github following command is used

$ git push -u origin master

```
PS C:\Users\HP 15L\Desktop\T21-79> git branch -m Main
PS C:\Users\HP 15L\Desktop\T21-79> git push -u origin Main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 24 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 657 bytes | 657.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'Main' on GitHub by visiting:
remote:      https://github.com/tsecpractical-star/SEPM_PRAC/pull/new/Main
remote:
To https://github.com/tsecpractical-star/SEPM_PRAC.git
 * [new branch]      Main -> Main
branch 'Main' set up to track 'origin/Main'.
PS C:\Users\HP 15L\Desktop\T21-79>
```

Now you can check the github for updated contents.



B) Pull Changes

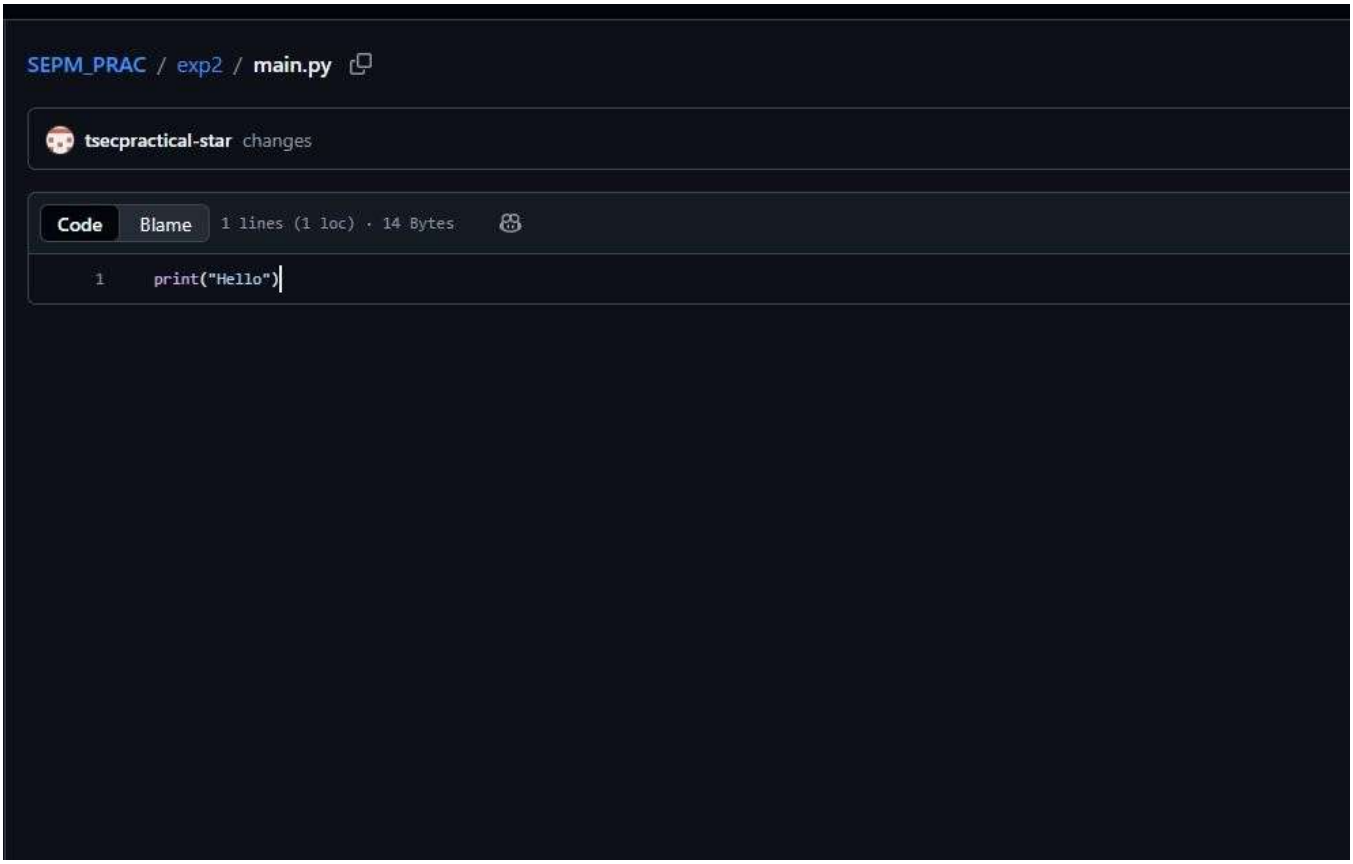
Pull command is used to download the remote updated repository into local one. The command for download is

```
$ git pull
```

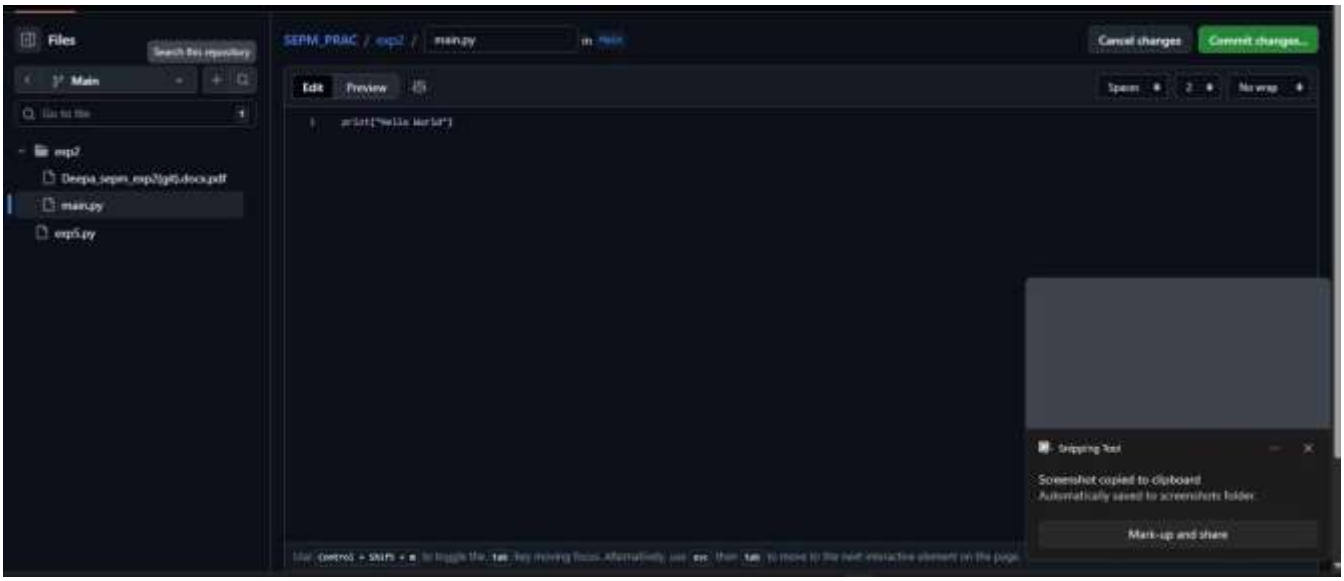
```
PS C:\Users\HP 15L\Desktop\T21-79> git pull
remote: Enumerating objects: 2, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 1.66 KiB | 113.00 KiB/s, done.
From https://github.com/tsecpractical-star/SEPM_PRAC
 * [new branch]      main       -> origin/main
Already up to date.
PS C:\Users\HP 15L\Desktop\T21-79>
```

C) Fetch

Suppose you have a file in github and you have changes that.



ORIGINAL FILE



Changed File

Now we use fetch command to fetch the changes, which will show you both the files like original and changed in local repository.

$ git fetch

```
PS C:\Users\HP 15L\Desktop\T21-79> git fetch
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 2.61 KiB | 178.00 KiB/s, done.
From https://github.com/tsecpractical-star/SEPM_PRAC
   011b9ae..09bf1dd  Main       -> origin/Main
 * [new branch]      main       -> origin/main
PS C:\Users\HP 15L\Desktop\T21-79>
```

Here fetch will not show you like updated changes file as like push. So use merge command to merge the changes so use following command for merge.

$ git merge origin/master

```
PS C:\Users\HP 15L\Desktop\T21-79> git merge origin/Main
>>
Merge made by the 'ort' strategy.
 exp5.py | 10 ----------
 1 file changed, 10 deletions(-)
 delete mode 100644 exp5.py
PS C:\Users\HP 15L\Desktop\T21-79>
Main  0↓ 2↑  ⊗ 0 ⚠ 0  0                Ln 1, Col 15  Spaces: 4  UTF-8
```

Online Resource: https://dzone.com/articles/top-20-git-commands-with-examples